Matteo Palmonari
matteo.palmonari@unimib.it

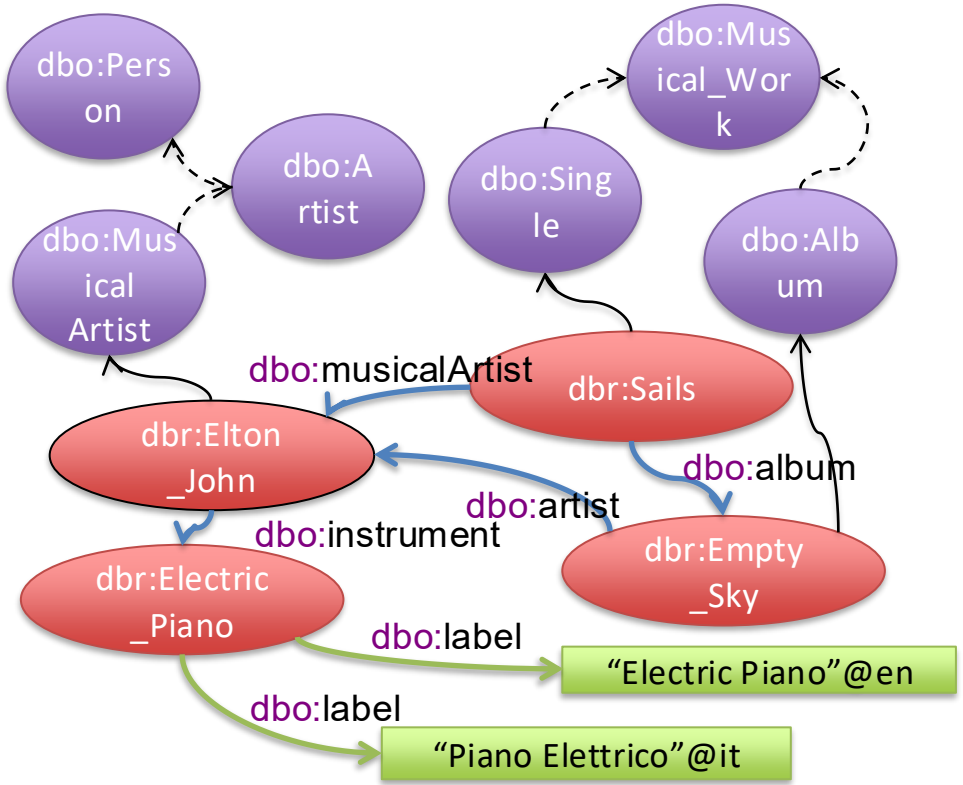
INSID&S Lab
***In**teraction and **S**emantics for*
***I**nnovation with **D**ata **&** **S**ervices*

2.4 Vocabularies for RDF

RDF: IRIfied Graphs and Triples

<dbr:Electric_Piano, rdfs:label, “Electric Piano”@en>
<dbr:Electric_Piano, rdfs:label, “Piano Elettrico”@it>
<dbr:Elton_John, dbo:instrument, dbr:Electric Piano>
<dbr:Empty_Sky, dbo:artist, dbr:Elton_John>
<dbr:Sails, dbo:musicalArtist, dbr:Elton_John>
<dbr:Sails, dbo:album, dbr:Empty_Sky>
<dbr:Elton_John, dbo:instrument, dbr:Electric_Piano>
<dbr:Elton_John, rdf:type, dbr:Musical_Artist>
<dbr:Sails, rdf:type, dbo:Single>
<dbr:Empty_Sky, rdf:type, dbo:Album>
<dbo:Musical_Artist, rdfs:subClassOf, dbo:Artist>
<dbo:Artist, rdfs:subClassOf, dbo:Person>
…

dbr: for http://dbpedia.org/resource/
e.g.: http://dbpedia.org/resource/Elton_John
dbo: for http://dbpedia.org/ontology/
rdfs for https://www.w3.org/2000/01/rdf-schema#
rdf for http://www.w3.org/1999/02/22-rdf-syntax-ns#



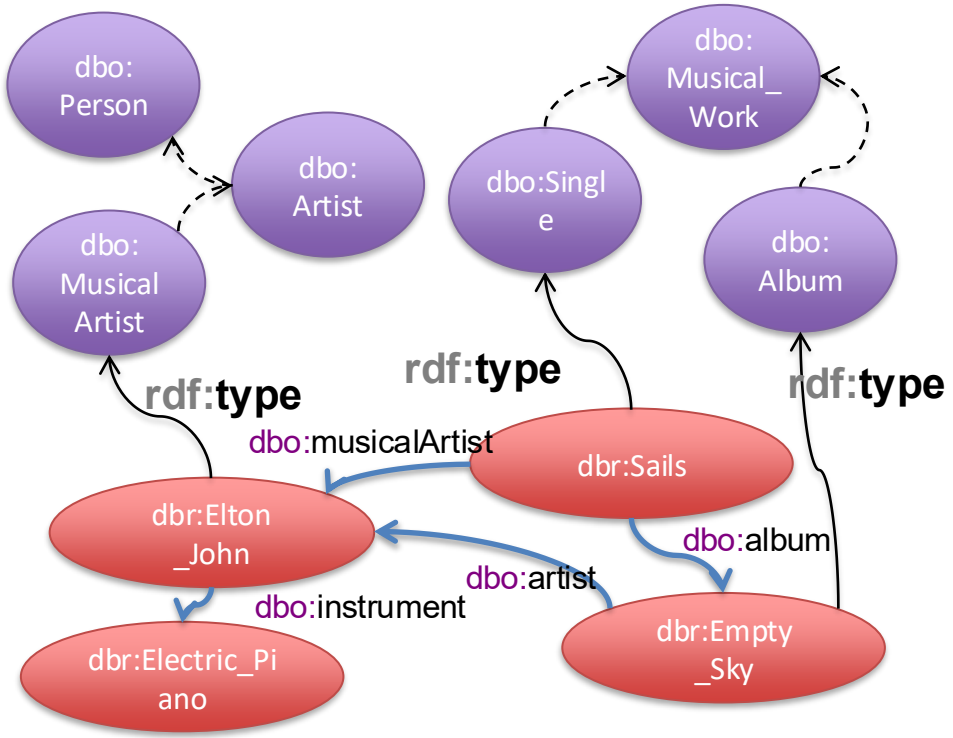
Categories by **rdf:type** assertions

Interpretation:
The resource in the *subject* belongs to the **category** specified by the *object*

Examples:

dbr:Elton_John **rdf:type** dbr:Musical_Artist .
dbr:Sails **rdf:type** dbo:Single .
dbr:Empty_Sky **rdf:type** dbo:Album .

…. one of the very few predicates defined in the RDF specification itself



Data Schemas: **RDBs** vs RDF

- Relational data (SQL): structured data
  - Structure is given by the database schema (defined with the DDL)
    - Tables (entities described in the DB), attributes and their domains, relations, constraints
  - Data and schema are **strongly coupled**
    - Schema-first: the schema is defined before entering the data
    - Prescriptive schema: data must be compliant with the schema to be inserted in the DB

```
CREATE TABLE Sells (
      bar         CHAR(20),
      beer        VARCHAR(20),
      price REAL,
      PRIMARY KEY (bar, beer)
);
```

Data Schemas: RDBs vs **RDF**

- RDF data: semi-structured data
  - Some structure is given by the data model (triples)
  - Data and schema are **loosly coupled**
    - Data must use URIs for properties and are expected to use URIs for classes → **vocabularies**
    - In principle… "data-first": data can be inserted without any schema definition, what is used is the vocabulary (not a good practice: remember, one of our goal is use terms with shared meaning)
    - Non-prescriptive "schemas": data are not required to be compliant with the schema to be inserted
      - Schema as "social agreement": community-based definitions of recommended terms and usage
      - Deductive schema: the schema defines the meaning of the terms in the vocabulary in terms of inferences (e.g., triples added to the KG, inconsistencies)

<dbr:Electric_Piano, rdfs:label, "Electric Piano
<dbr:Electric_Piano, rdfs:label, "Piano Elettric
<dbr:Elton_John, dbo:instrument, dbr:Electri
<dbr:Empty_Sky, dbo:artist, dbr:Elton_John>
<dbr:Sails, dbo:musicalArtist, dbr:Elton_Johr
<dbr:Sails, dbo:album, dbr:Empty_Sky>
<dbr:Elton_John, dbo:instrument, dbr:Electri
<dbr:Elton_John, rdf:type, dbr:Musical_Artist
<dbr:Sails, rdf:type, dbo:Single>
<dbr:Empty_Sky, rdf:type, dbo:Album>
<dbo:Musical_Artist, rdfs:subClassOf, dbo:A
<dbo:Artist, rdfs:subClassOf, dbo:Person>

…

dbr: for http://dbpedia.org/resource/
e.g.: http://dbpedia.org/resource/Elton
dbo: for http://dbpedia.org/ontology/
rdfs for https://www.w3.org/2000/01/rdf-schem
rdf for http://www.w3.org/1999/02/22-rdf-synta

Often referred to as "ontologies"

Logic-based languages RDFS/OWL

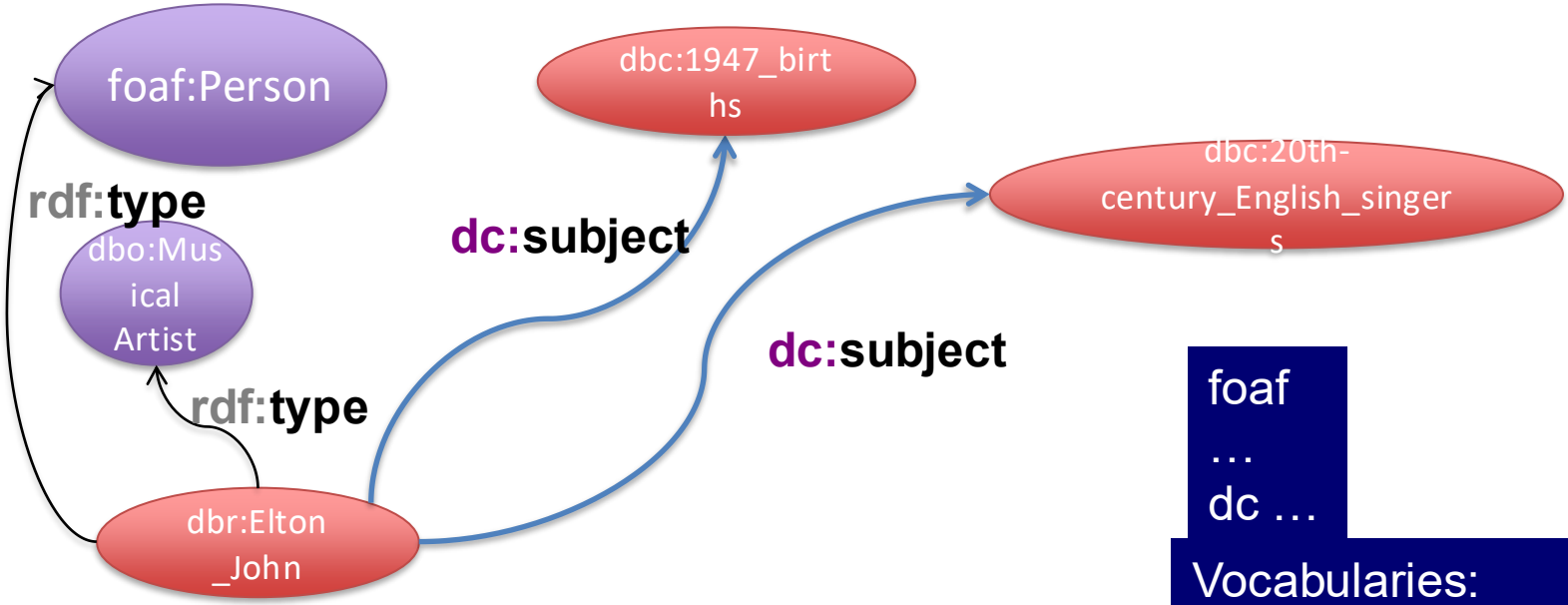
Data Schemas: RDBs vs RDF with Example

- Databases
  - Interpretation of schemas (example)
    - A *constraint-oriented* schema
      - Song(ID,Title,*Artist*,Year)
      - Artist(ID,Name,Label)
      - Values for the attribute Artist in the Song table must be an (existing) ID in the Artist table
      - Name in Artist must be UNIQUE
    - Implications
      - If the ID does not exist in the Artist table, data cannot be entered in the Song table
      - Information in the Artist table must be filled in completely (or NULL for non-key attributes)
      - If a name is taken, it cannot be reused (e.g., “Placebo (2)”)

- RDF
  - Interpretation of schemas (example)
    - A *deductive* schema (simplified so far)
      - Songs are artworks
      - Songs’ artists must be of type MusicArtist
      - Values of year must be of type xsd:Year
      - MusicArtists are Agents
    - Implications
      - If **ex:s042** is a song, then it is an Artwork
      - If **ex:a5946** is artist of some song, then it is a MusicArtist, and also an Agent
      - If 2023 is a year of some song, then it is a xsd:Year
      - In any case we can add a triple if it does not violate RDF model rules
      - A contradiction can be generated if conflicting information is given, e.g., if **ex:a5946** is a Country, and countries are defined to be not Agensts

Constraints in terms of syntactic patterns and violations not part of deductive schema; can be defined/validated using SHACL Shapes Constraint Language

Shared Vocabularies with RDF



foaf
…
dc …

Vocabularies:
- Properties
- Categories

…with shared meaning

RDF in Action: FOAF

The *Friend of a Friend* (FOAF) project is creating a Web of machine-readable pages describing people, the links between them and the things they create and do

```
@prefix foaf: <http://purl.org/dc/elements/1.1/> .
<foaf:Person rdf:about="#danbri" xmlns:foaf="http://xmlns.com/foaf/0.1/">
<foaf:name>Dan Brickley</foaf:name>
<foaf:homepage rdf:resource="http://danbri.org/" />
<foaf:openid rdf:resource="http://danbri.org/" />
<foaf:img rdf:resource="/images/me.jpg" />
</foaf:Person>
```

example of FOAF

metadata describing Dan Brickley

http://www.foaf-project.org/
http://xmlns.com/foaf/spec/

FOAF at a glance

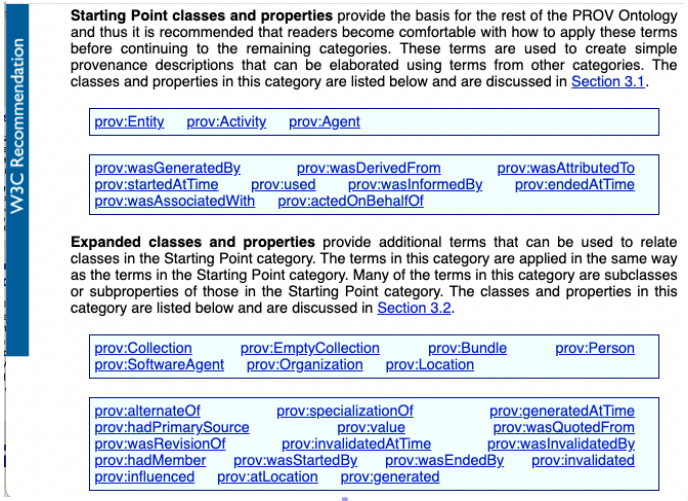
An a-z index of FOAF terms, by class (categories or types) and by property.

Classes: | Agent | Document | Group | Image | LabelProperty | OnlineAccount | OnlineChatAccount | OnlineEcommerceAccount | OnlineGamingAccount | Organization | Person | PersonalProfileDocument | Project |

Properties: | account | accountName | accountServiceHomepage | age | aimChatID | based_near | birthday | currentProject | depiction | depicts | dnaChecksum | familyName | family_name | firstName | fundedBy | geekcode | gender | givenName | givenname | holdsAccount | homepage | icqChatID | img | interest | isPrimaryTopicOf | jabberID | knows | lastName | logo | made | maker | mbox | mbox_sha1sum | member | membershipClass | msnChatID | myersBriggs | name | nick | openid | page | pastProject | phone | plan | primaryTopic | publications | schoolHomepage | sha1 | skypeID | status | surname | theme | thumbnail | tipjar | title | topic | topic_interest | weblog | workInfoHomepage | workplaceHomepage | yahooChatID |

PROV-O: The Provenance Vocabulary

https://www.w3.org/TR/prov-o/



W3C Recommendation

Starting Point classes and properties provide the basis for the rest of the PROV Ontology and thus it is recommended that readers become comfortable with how to apply these terms before continuing to the remaining categories. These terms are used to create simple provenance descriptions that can be elaborated using terms from other categories. The classes and properties in this category are listed below and are discussed in Section 3.1.

prov:Entity prov:Activity prov:Agent

prov:wasGeneratedBy prov:wasDerivedFrom prov:wasAttributedTo
prov:startedAtTime prov:used prov:wasInformedBy prov:endedAtTime
prov:wasAssociatedWith prov:actedOnBehalfOf

Expanded classes and properties provide additional terms that can be used to relate classes in the Starting Point category. The terms in this category are applied in the same way as the terms in the Starting Point category. Many of the terms in this category are subclasses or subproperties of those in the Starting Point category. The classes and properties in this category are listed below and are discussed in Section 3.2.

prov:Collection prov:EmptyCollection prov:Bundle prov:Person
prov:SoftwareAgent prov:Organization prov:Location

prov:alternateOf prov:specializationOf prov:generatedAtTime
prov:hadPrimarySource prov:value prov:wasQuotedFrom
prov:wasRevisionOf prov:invalidatedAtTime prov:wasInvalidatedBy
prov:hadMember prov:wasStartedBy prov:wasEndedBy prov:invalidated
prov:influenced prov:atLocation prov:generated

As used at Amazon…

It does scale! Here is a great example…

Inventory knowledge graph of Amazon Fulfillment is used to

- investigate fulfillment processes
- investigate lost & found -issues
- improve precision of product recalls

Extends the PROV-O ontology: models the end-to-end logistics process as a form of provenance

Runs on multiple (federated) Neptune clusters

Size:

1T+ triples

4B new triples per day

Queries (p95):

< 50 ms to find a node

< 1 s to retrieve the whole path

From Ora Lassila's keynote at ISWC 2024
Principal Technologist in the Amazon Neptune graph database team and the co-chair of the W3C RDF-star Working Group

Vocabularies as Social Agreements

- A group of persons can agree to use a specific **vocabulary**; in RDF:
  - **Classes** to type resources, e.g., foaf:Document
  - **Properties** to describe resources, e.g., foaf:topic
  - **Prescritpions / recommendations** on how to use the properties:
    - which types of subjects, e.g. use foaf:topic to describe resources of type foaf:Document or foaf:Image
    - which types of objects, e.g., the objects for foaf:made should not be a literal

Schema.org

- “a collaborative, community activity with a mission to create, maintain, and promote schemas for structured data on the Internet, on web pages, in email messages, and beyond.”
  - can be used with many different encodings, including RDFa, Microdata and JSON-LD
  - 10+M websites using it
  - Collaborative effort by big internet players (Google, Microsoft, Yahoo and Yandex)
  - Evolving along time

- Visit and browse at: https://schema.org/

Example - Schema.org:Event

Event

Thing > Event

An event happening at a certain time and location, such as a concert, lecture, or festival. Ticketing information may be added via the offers property. Repeated events may be structured as separate Event objects.

[more...]

Property	Expected Type	Description
Properties from Event		
about	Thing	The subject matter of the content. Inverse property: subjectOf.
actor	Person	An actor, e.g. in tv, radio, movie, video games etc., or in an event. Actors can be associated with individual items or with a series, episode, clip. Supersedes actors.
aggregateRating	AggregateRating	The overall rating, based on a collection of reviews or ratings, of the item.
attendee	Organization or Person	A person or organization attending the event. Supersedes attendees.
audience	Audience	An intended audience, i.e. a group for whom something was created. Supersedes serviceAudience.
composer	Organization or Person	The person or organization who wrote a composition, or who is the composer of a work performed at some event.

FOAF Primitives

What is a Person?

FOAF terms, grouped in broad categories.

FOAF Basics

- Agent
- Person
- name
- nick
- title
- homepage
- mbox
- mbox_sha1sum
- img
- depiction (depicts)
- surname
- familyName
- givenName
- firstName
- lastName

Personal Info

- weblog
- knows
- interest
- currentProject
- pastProject
- plan
- based_near
- age
- workplaceHomepage
- workInfoHomepage
- schoolHomepage
- topic_interest
- publications
- geekcode
- myersBriggs
- dnaChecksum

Online Accounts / IM

- OnlineAccount
- OnlineChatAccount
- OnlineEcommerceAccount
- OnlineGamingAccount
- account
- accountServiceHomepage
- accountName
- icqChatID
- msnChatID
- aimChatID
- jabberID
- yahooChatID
- skypeID

How can we specify the meaning of types and predicates?

A **vocabulary** (a list of allowed terms) is not enough…
In **Ontologies** we use axioms to constrain the meaning of terms in a way that is useful for a machine, i.e., in a way that even a machin can make inferences on top of the data

Ontologies

ontology noun

Save Word

on·tol·o·gy | \ än-ˈtä-lə-jē \

Definition of *ontology*

1 : a branch of metaphysics concerned with the nature and relations of being
// Ontology deals with abstract entities.

2 : a particular theory about the nature of being or the kinds of things that have existence

Merriam-Webster

Ontologies in Information Sciences by Tom Gruber

- An ontology is a description (like a formal specification of a program) of the **concepts** and **relationships** that can formally exist **for** an agent or **a community of agents**. This definition is consistent with the usage of ontology as set of concept definitions, but more general. And it is a different sense of the word than its use in philosophy.

- […] To specify a conceptualization, one needs to state axioms that do constrain the possible interpretations for the defined terms**

* Gruber, T. (2001). "What is an Ontology?". Stanford University.
** Gruber, T (1993). "A translation approach to portable ontology specifications". Knowledge Acquisition. 5 (2): 199–220.

(Axiomatic) Ontologies

An axiomatic ontology is a formal specification of a conceptualization by means of **a language *L***, a set of **logical axioms of L** and a formal semantics that uniquely determines the **meaning** of the terms, i.e., the symbols of *L*.

To determine the meaning of a term → to specify its unambiguous interpretation within a mathematical model

Axiomatic ontologies organize knowledge by defining:

- Symbols to represent **individuals** (e.g., entitites like *Elton John*), **classes** (e.g., entity types like *Musical Artist*), **relations** (e.g., properties to describe entities like *made*) – the vocabulary
- Axioms to represent the logical constraints that specify their meaning and thus support **automated reasoning** (**inference**)

Reminder: Reasoning and Inference

rea·son·ing | \ ˈrēz-niŋ, ˈrē-zᵊn-iŋ \

Definition of *reasoning*

1 : the use of reason

especially : the drawing of inferences or conclusions through the use of reason

2 : an instance of the use of reason : ARGUMENT

- Inferences / conclusions
  - An inference establishes a relation
    - Premises → Conclusion
  - Computational models of inference
    - Which conclusions can we draw from a set of premises?
    - Why is the conclusion correct?

Reminder: Reasoning and Inference

- Goal: model agents that can manipulate knowledge and execute knowledge-intensive tasks
  - Facts + general domain knowledge
    - {Paris is located in France, France is located in Europe} + {is located in is a transitive relation} ⊆ KB
  - Reasoning: inferences and inference-aware queries
    - KB ⊢ {Paris is located in Europe}
    - Where is Paris located? {France, Europe}
    - … more on this when we introduce ontologies ☺

Some Remarks on Schemas/Ontologies

- More on: ontologies vs database schemas
  - Database schemas: several technical details (e.g., length of strings)
  - Ontologies: modeling is more conceptual (e.g., all musicians are persons, persons cannot be friend of places, etc.)
  - Technical data modeling for RDF: Shapes Constraint Language (SHACL)
    - W3C recommendation from 2017: https://www.w3.org/TR/shacl/
    - Detailed constraints on the data
    - Specific engines can validate the data (data quality)
- Property-graph data model
  - Also: semi-structured, loose schema/data coupling
  - Usually schema is somehow defined: node types, relationship types, etc.

Recap

- RDF: a data model for publishing data on the web
  - Triples using IRIs + literals + blank nodes
- RDF KGs: “linked data”, optimized for knowledge sharing on the web
- Vocabularies: organization of knowledge to support sharing and interoperability
- Highly recommended: explore DBpedia and Wikidata